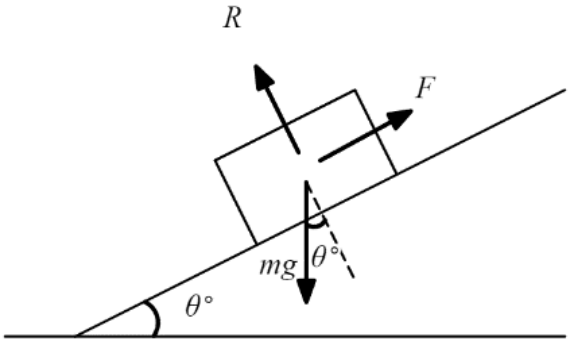
2.55 Forces: Inclined Planes

Worked Solutions

1 a)



1 b)

$R(\nwarrow\!\!\searrow)$: $R - 5 \times 9.8\cos 30 = 0 \Rightarrow R = \frac{49}{2}\sqrt{3}$

$R(\nearrow)$: $F - 5 \times 9.8\sin 30 = 0 \Rightarrow F = \frac{49}{2}$

$\frac{49}{2} = \mu \times \frac{49}{2}\sqrt{3} \Rightarrow \mu = 0.577$ (3 SF)

$\mu = 0.58$ (2 SF – AQA)

1 c)

$R(\nwarrow\!\!\searrow)$: $R - 5 \times 9.8\cos 45 = 0 \Rightarrow R = \frac{49}{2}\sqrt{2}$

$R(\nearrow)$: $F - 5 \times 9.8\sin 45 = 0 \Rightarrow F = \frac{49}{2}\sqrt{2}$

$\frac{49}{2}\sqrt{2} = \mu \times \frac{49}{2}\sqrt{2} \Rightarrow \mu = 1$

1 d)

$R(\nwarrow\!\!\searrow)$: $R - 5 \times 9.8\cos 55 = 0 \Rightarrow R = 49\cos 55$

$R(\nearrow)$: $F - 5 \times 9.8\sin 55 = 0 \Rightarrow F = 49\sin 55$

$49\sin 55 = \mu \times 49\cos 55 \Rightarrow \mu = 1.43$ (3 SF)

$\mu = 1.4$ (2 SF – AQA)

1 e)

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8\cos\theta = 0 \Rightarrow R = 9.8m\cos\theta$

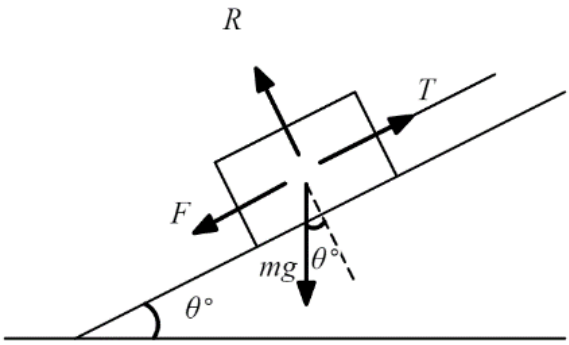
$R(\nearrow)$: $F - m \times 9.8\sin\theta = 0 \Rightarrow F = 9.8m\sin\theta$

$9.8m\sin\theta = \mu \times 9.8m\cos\theta$

$\sin\theta = \mu \times \cos\theta$

$\Rightarrow \mu = \tan\theta$ as req.

2 a)



2 b)

$R(\nwarrow\!\!\searrow)$: $R - 4 \times 9.8\cos 30 = 0 \Rightarrow R = \frac{98}{5}\sqrt{3}$

$F = 0.8 \times \frac{98}{5}\sqrt{3}$

$F = 27.2$ (3 SF)

$F = 27$ (2 SF – AQA)

$R(\nearrow)$: $T - 0.8 \times \frac{98}{5}\sqrt{3} - 4 \times 9.8\sin 30 = 0$

$T = 46.8$ (3 SF)

$T = 47$ (2 SF – AQA)

2 c)

$R(\nwarrow\!\!\searrow)$: $R - 5 \times 9.8\cos 60 = 0 \Rightarrow R = 24.5$

$F = 0.6 \times 24.5 = 14.7$

$F = 15$ (2 SF – AQA)

$R(\nearrow)$: $T - 14.7 - 5 \times 9.8\sin 60 = 0$

$T = 57.1$ (3 SF)

$T = 57$ (2 SF – AQA)

2 d)

$R(\nearrow)$: $10 - 5 - m \times 9.8\sin 20 = 0$

$m = 1.49$ (3 SF)

$m = 1.5$ (2 SF – AQA)

2 e)

$R(\nearrow)$: $12 - 8 - 2 \times 9.8\sin\theta = 0$

$\theta = 11.8$ (3 SF)

$\theta = 12$ (2 SF – AQA)

2 f)

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8\sin 45 = 0$

$F = 0.6 \times \frac{49}{10}\sqrt{2}m$

$R(\nearrow)$: $10 - 0.6 \times \frac{49}{10}\sqrt{2}m - m \times 9.8 \sin 45 = 0$

$\therefore 10 = (0.6 \times \frac{49}{10}\sqrt{2} + \frac{49}{10}\sqrt{2})m$

$m = 0.902$ (3 SF)

$m = 0.90$ (2 SF – AQA)

2 g)

$R(\nwarrow)$: $R - mg\cos\theta = 0$

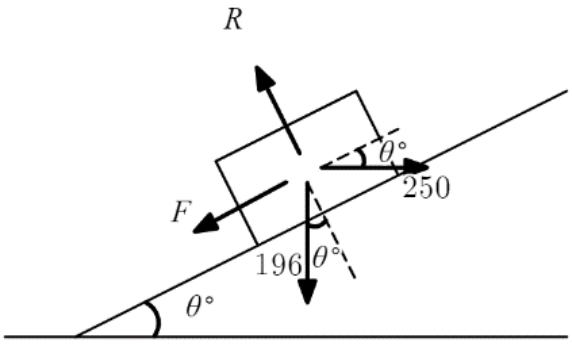
$F = \mu mg\cos\theta$

$R(\nearrow)$: $T - \mu mg\cos\theta - mg\sin\theta = 0$

$T = \mu mg\cos\theta + mg\sin\theta$

$T = mg(\mu\cos\theta + \sin\theta)$

3 a)



3 b)

$\sqrt{3^2 + 4^2} = 5$

Hence $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

3 c)

$R(\nwarrow)$: $R - 196 \times \frac{4}{5} - 250 \times \frac{3}{5} = 0 \Rightarrow R = 306.8$ N

$R = 310$ N (2 SF – AQA)

3 d)

$R(\nearrow)$: $250 \times \frac{4}{5} - F - 196 \times \frac{3}{5} = 0 \Rightarrow F = 82.4$ N

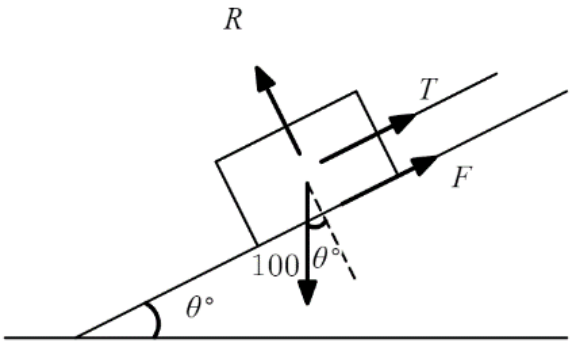
$F = 82$ N (2 SF – AQA)

3 e)

$82.4 = 306.8\mu \Rightarrow \mu = 0.269$ (3 SF)

$\mu = 0.27$ (2 SF – AQA)

4 a)



4 b)

$\sqrt{7^2+24^2}=25 \Rightarrow \sin\theta=\frac{7}{25}$ and $\cos\theta=\frac{24}{25}$

4 c)

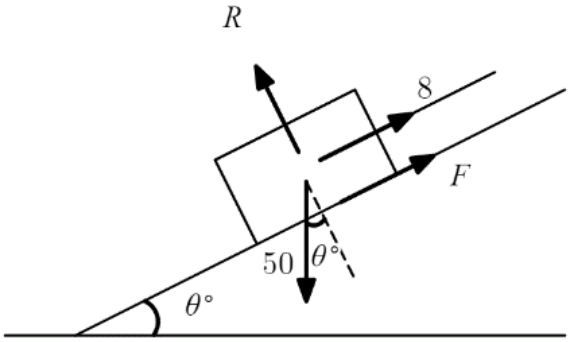
$R(\nwarrow\!\!\searrow)$: $R-100\times\frac{24}{25}=0 \Rightarrow R=96$

$F=0.25\times 96=24$

$R(\nearrow)$: $T+24-100\times\frac{7}{25}=0$

$T=4$

5 a)



5 b)

$\sqrt{2^2+25^2}=\sqrt{29} \Rightarrow \sin\theta=\frac{2}{\sqrt{29}}$ and $\cos\theta=\frac{5}{\sqrt{29}}$

5 c)

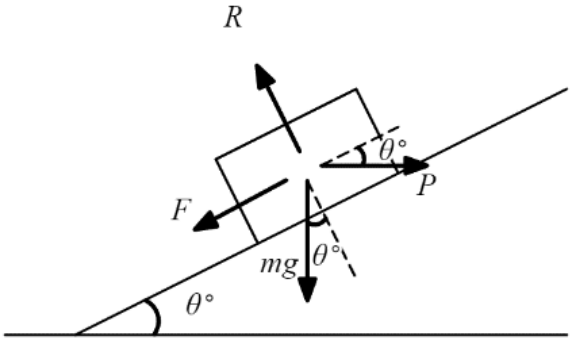
$R(\nwarrow\!\!\searrow)$: $R-50\times\frac{5}{\sqrt{29}}=0 \Rightarrow R=\frac{250}{\sqrt{29}}$

$R(\nearrow)$: $8+F-50\times\frac{2}{\sqrt{29}}=0 \Rightarrow F=\frac{100}{\sqrt{29}}-8$

$(\frac{100}{\sqrt{29}}-8)=\frac{250}{\sqrt{29}}\mu \Rightarrow \mu=0.228$ (3 SF)

$\mu=0.2$ (1 SF – AQA)

6 a)



6 b)

$R(\nwarrow\!\searrow)$: $R - 2 \times 10\cos 30 - P\sin 30 = 0$

$R = \frac{1}{2}P + 10\sqrt{3}$ N

6 c)

$F = 0.4 \times (\frac{1}{2}P + 10\sqrt{3}) = \frac{1}{5}P + 4\sqrt{3}$

6 d)

$R(\nearrow)$: $P\cos 30 - (\frac{1}{5}P + 4\sqrt{3}) - 2 \times 10\sin 30 = 0$

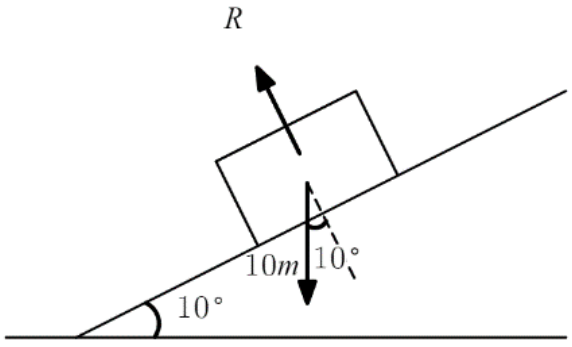
$\frac{\sqrt{3}}{2}P - \frac{1}{5}P - 4\sqrt{3} - 10 = 0$

$(\frac{\sqrt{3}}{2} - \frac{1}{5})P = 4\sqrt{3} + 10$

$P = 25.1$ (3 SF)

$P = 30$ (1 SF – AQA)

7 a)



7 b)

$R(\swarrow\!\nearrow)$: $10m\sin 10 = ma$

$\Rightarrow a = 1.736481777\ldots$ down the slope.

$a = 1.74$ ms^{-2} down the slope (3 SF)

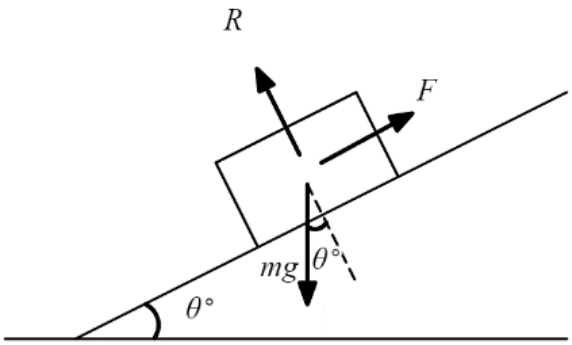
$a = 2$ ms^{-2} down the slope (1 SF – AQA)

7 c)

$v^2 = 0^2 + 2 \times 10 \sin 10 \times 5 \Rightarrow v = 4.17 \text{ ms}^{-1}$ (3 SF)

$v = 4 \text{ ms}^{-1}$ (1 SF – AQA)

8 a)



8 b)

$R(\nwarrow\!\!\searrow)$: $R - 4 \times 10 \cos 60 = 0 \Rightarrow R = 20$

$F = 0.2 \times 20 = 4$

$R(\swarrow\!\!\nearrow)$: $4 \times 10 \sin 60 - 4 = 4a \Rightarrow a = 7.66$

$\therefore a = 7.66 \text{ ms}^{-2}$ down the slope (3 SF)

$a = 8 \text{ ms}^{-2}$ down the slope (1 SF – AQA)

8 c)

$R(\swarrow\!\!\nearrow)$: $12 \times 10 \sin 45 - F = 12 \times 1.5$

$\Rightarrow F = 60\sqrt{2} - 18$

$R(\nwarrow\!\!\searrow)$: $R - 12 \times 10 \cos 45 = 0 \Rightarrow R = 60\sqrt{2}$

$60\sqrt{2} - 18 = 60\sqrt{2}\mu \Rightarrow \mu = 0.789$ (3 SF)

8 d)

$R(\nwarrow\!\!\searrow)$: $R - mg \cos\theta = 0 \Rightarrow R = mg \cos\theta$

$F = \mu mg \cos\theta$

$R(\swarrow\!\!\nearrow)$: $mg \sin\theta - \mu mg \cos\theta = ma$

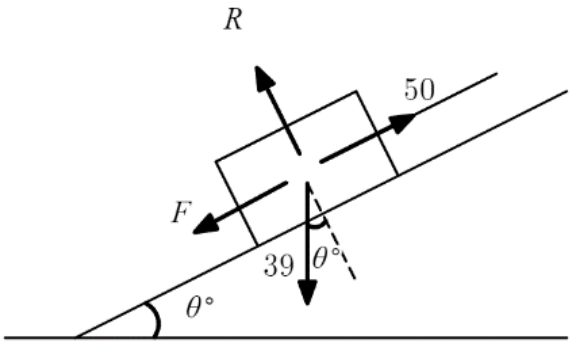
$g \sin\theta - \mu g \cos\theta = a$

$\therefore a = g(\sin\theta - \mu \cos\theta)$ as req.

8 e)

$a = g \sin\theta$

9 a)



9 b)

$R(\nwarrow\!\!\searrow)$: $R - 39 \times \frac{12}{13} = 0 \Rightarrow R = 36$

$F = 0.5 \times 36 = 18$

$R(\nearrow)$: $50 - 18 - 39 \times \frac{5}{13} = 3.9a \Rightarrow a = \frac{170}{39}$ as req.

9 c)

$s = 0 + \frac{1}{2} \times \frac{170}{39} \times 1.5^2 = \frac{255}{52} = 4.90$ m (3 SF)

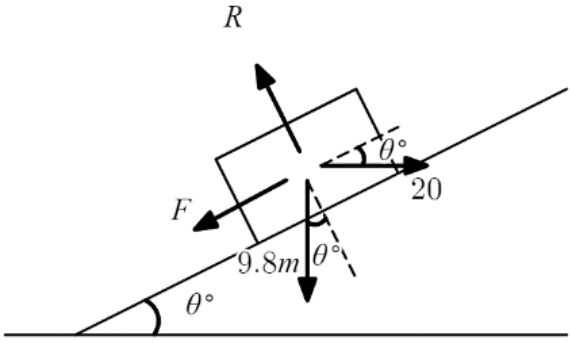
$s = 5$ m (1 SF – AQA)

9 d)

$v = 0 + \frac{170}{39} \times 1.5 = \frac{85}{13} = 6.54$ ms^{-1} (3 SF)

$v = 7$ ms^{-1} (1 SF – AQA)

10 a)



10 b)

$\sqrt{7^2 + 3^2} = \sqrt{58} \Rightarrow \cos\theta = \frac{7}{\sqrt{58}}$ and $\sin\theta = \frac{3}{\sqrt{58}}$

10 c)

$R(\nwarrow\!\!\searrow)$: $R - 9.8 \times 10 \times \frac{7}{\sqrt{58}} - 20 \times \frac{3}{\sqrt{58}} = 0 \Rightarrow R = \frac{746}{\sqrt{58}}$

$F = 0.25 \times \frac{746}{\sqrt{58}} = \frac{373}{2\sqrt{58}}$

$R(\nearrow)$: $20 \times \frac{7}{\sqrt{58}} - \frac{373}{2\sqrt{58}} - 9.8 \times 10 \times \frac{3}{\sqrt{58}} = 10a$

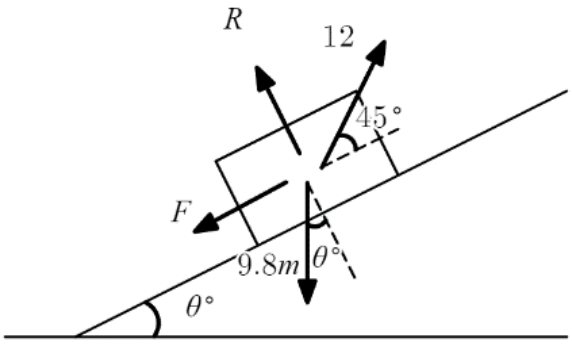
$a = -4.47$ ms^{-2} (3 SF)

$a = -4.8$ ms^{-2} (2 SF – AQA)

10 d)

Deceleration up the slope.

11 a)



11 b)

$\sqrt{7^2 + 24^2} = 25 \Rightarrow \cos\theta = \frac{24}{25}$ and $\sin\theta = \frac{7}{25}$

11 c)

$R(\nwarrow)$: $R - 2 \times 9.8 \times \frac{24}{25} + 12 \times \frac{\sqrt{2}}{2} = 0$

$\Rightarrow R = 10.33071863$

$R(\nearrow)$: $12 \times \frac{\sqrt{2}}{2} - F - 2 \times 9.8 \times \frac{7}{25} = 1.2 \times 2$

$\Rightarrow F = 0.5972813742\ldots$

$0.5972813742\ldots = 10.33071863\ldots\mu \Rightarrow$

$\mu = 0.0578$ (3 SF)

$\mu = 0.058$ (2 SF – AQA)

11 d)

$R(\nwarrow)$: $R - m \times 9.8 \times \frac{24}{25} + 12 \times \frac{\sqrt{2}}{2} = 0$

$\Rightarrow R = \frac{1176}{125}m - 6\sqrt{2}$

$F = 0.1(\frac{1176}{125}m - 6\sqrt{2}) = \frac{588}{625}m - \frac{3}{5}\sqrt{2}$

$R(\nearrow)$: $12 \times \frac{\sqrt{2}}{2} - \frac{588}{625}m + \frac{3}{5}\sqrt{2} - m \times 9.8 \times \frac{7}{25} = 1.2m$

$12 \times \frac{\sqrt{2}}{2} - \frac{588}{625}m + \frac{3}{5}\sqrt{2} = \frac{493}{125}m$

$\Rightarrow m = 1.91$ (3 SF)

$m = 1.9$ (2 SF – AQA)

12 a)

$R(\nwarrow\!\!\searrow)$: $R - 6 \times 9.8\cos 40 = 0 \Rightarrow R = 58.8\cos 40$

$F = 0.45 \times 58.8\cos 40 = 26.46\cos 40$

$R(\nearrow)$: $25 - 26.46\cos 40 - 6 \times 9.8\sin 40 = 6a$

$\Rightarrow a = -5.51$ ms^{-2} (3 SF)

$a = -5.5$ ms^{-2} (2 SF – AQA)

12 b)

It is decelerating up the slope.

12 c)

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8\cos 40 = 0 \Rightarrow R = 9.8m\cos 40$

$F = 0.45 \times 9.8m\cos 40 = 4.41m\cos 40$

$R(\swarrow\!\!\nearrow)$: $-4.41m\cos 40 + 9.8m\sin 40 = ma$

$-4.41\cos 40 + 9.8\sin 40 = a$

$\Rightarrow a = 2.921062581\ldots$

$a = 2.92$ ms^{-2} (3 SF)

$a = 2.9$ ms^{-2} (2 SF – AQA)

13 a)

$R(\nwarrow\!\!\searrow)$: $R - 4 \times 9.8\cos 24 = 0 \Rightarrow R = 39.2\cos 24$

$R(\nearrow)$: $35 - F - 4 \times 9.8\sin 24 = 4 \times 0.5$

$\Rightarrow F = 33 - 39.2\sin 24$

$33 - 39.2\sin 24 = \mu(39.2\cos 24)$

$\Rightarrow \mu = 0.4762763451$ so $\mu = 0.48$ as req

13 b)

$2 = 0 + \frac{1}{2} \times 0.5t^2 \Rightarrow t = 2\sqrt{2} = 2.83$ seconds

$t = 2.83$ s (3 SF)

$t = 2.8$ s (2 SF – AQA)

13 c)

Initial speed given by final speed of first phase:

$v^2 = 0 + 2 \times 0.5 \times 2 \Rightarrow v = \sqrt{2}$

New acceleration:

$R(\nwarrow\!\!\searrow)$: $R - 39.2\cos 24 + 35\sin 20 = 0$

$F = 0.48(39.2\cos 24 - 35\sin 20) = 11.44333292\ldots$

$R(\nearrow)$: $35\cos 20 - 11.44\ldots - 39.2\sin 24 = 4a$

$\Rightarrow a = 1.3754581\ldots$

$2 = \sqrt{2}t + \frac{1}{2} \times 1.3754581\ldots \times t^2$

$\Rightarrow t = 0.96312\ldots$ or $t = -3.01947\ldots$

So $t = 0.963$ seconds (3 SF)

$t = 0.96$ seconds (2 SF – AQA)

14 a)

$5 = 0 + \frac{1}{2}a \times 2^2 \Rightarrow a = 2.5$ ms^{-2}

14 b)

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8 \times \frac{4}{5} = 0 \Rightarrow R = \frac{196}{25}m$

$F = \frac{196}{25}m\mu$

$R(\swarrow\!\!\nearrow)$: $-\frac{196}{25}m\mu + m \times 9.8 \times \frac{3}{5} = 2.5m$

$-\frac{196}{25}\mu + \frac{147}{25} = 2.5 \Rightarrow \mu = \frac{169}{392} = 0.431$ (3 SF)

$\mu = 0.43$ (2 SF – AQA)

15)

$\sin\theta = \frac{12}{13}, \cos\theta = \frac{5}{13}$

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8 \times \frac{5}{13} = 0 \Rightarrow R = \frac{49}{13}m$

$F = 0.8 \times \frac{49}{13}m = \frac{196}{65}m$

$R(\nearrow\!\!\swarrow)$: $-m \times 9.8 \times \frac{12}{13} - \frac{196}{65}m = ma$

$\Rightarrow a = -\frac{784}{65}$

$0 = 20t + \frac{1}{2} \times -\frac{784}{65} \times t^2 \Rightarrow t = \frac{325}{98} = 3.32$ s (3 SF)

$t = 3.3$ s (2 SF – AQA)

16)

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

$0 = 15 \times 3 + \frac{1}{2} \times a \times 9 \Rightarrow a = -10$ ms^{-2}

$R(\nearrow\!\!\swarrow)$: $-m \times 9.8 \times \frac{3}{5} - F = -10m \Rightarrow F = \frac{103}{25}m$

$R(\nwarrow\!\!\searrow)$: $R - m \times 9.8 \times \frac{4}{5} = 0 \Rightarrow R = \frac{196}{25}m$

$\frac{103}{25}m = \mu \times \frac{196}{25}m \Rightarrow \mu = \frac{103}{196} = 0.526$ (3 SF)

$\mu = 0.53$ (2 SF – AQA)

17 a)

$R(\nwarrow\!\!\searrow)$: $15\sqrt{3} - 20\cos 30 - P\sin 30 = 0 \Rightarrow P = 10\sqrt{3}$

$R(\nearrow\!\!\swarrow)$: $10\sqrt{3}\cos 30 - F - 20\sin 30 = \frac{100}{49} \times -2.45$

$\Rightarrow F = 10$

17 b)

$10 = \mu \times 15\sqrt{3} \Rightarrow \mu = \frac{2}{9}\sqrt{3}$

$R(\nwarrow\!\!\searrow)$: $R - 20\cos 30 = 0 \Rightarrow R = 10\sqrt{3}$

$F = \frac{2}{9}\sqrt{3} \times 10\sqrt{3} = \frac{20}{3}$

$R(\swarrow\!\!\nearrow)$: $-\frac{20}{3} + 20\sin 30 = \frac{100}{49}a$

$\Rightarrow a = \frac{49}{30} = 1.63$ ms^{-2} down the slope (3 SF)

$\Rightarrow a = 1.6$ ms^{-2} down the slope (2 SF – AQA)

17 c)

Do not model the block as a particle, ie consider dimensions. Include air resistance. Consider any rotational effects of forces such as spin.

18 a)

$\sqrt{5^2 + 12^2} = 13 \Rightarrow \cos\theta = \frac{12}{13}$ and $\sin\theta = \frac{5}{13}$

$R(\nwarrow)$: $R - 49 \times \frac{12}{13} = 0 \Rightarrow R = \frac{588}{13}$

$R(\nearrow)$: $\frac{314}{13} - F - 49 \times \frac{5}{13} = 5 \times -1.2 \Rightarrow F = \frac{147}{13}$

$\frac{147}{13} = \frac{588}{13}\mu \Rightarrow \mu = 0.25$

18 b)

$s = 0 - \frac{1}{2} \times -1.2 \times 4^2 = 9.6$ m

18 c)

$F = 0.25 \times \frac{588}{13} = \frac{147}{13}$

$R(\swarrow)$: $-\frac{147}{13} + 49 \times \frac{5}{13} = \frac{49}{9.8}a$

$a = \frac{98}{65}$

$9.6 = 0 + \frac{1}{2} \times \frac{98}{65} \times t^2 \Rightarrow t = 3.57$ s (3 SF)

$t = 3.6$ s (2 SF – AQA)

19 a)

$R(\nwarrow)$: $R - mg \times \frac{3}{5} - P \times \frac{4}{5} = 0$

$\Rightarrow R = \frac{3}{5}mg + \frac{4}{5}P$

$\Rightarrow F = 0.6(\frac{3}{5}mg + \frac{4}{5}P)$

$R(\nearrow)$: $\frac{3}{5}P - \frac{4}{5}mg - \frac{9}{25}mg - \frac{12}{25}P \geq 0$

$\frac{3}{25}P \geq \frac{29}{25}mg$

$P \geq \frac{29}{3}mg$

19 b)

$R(\nearrow)$: $\frac{3}{5}Q - \frac{4}{5}mg - \mu(\frac{3}{5}mg + \frac{4}{5}Q) \leq 0$

$\frac{3}{5}Q - \frac{4}{5}mg \leq \mu(\frac{3}{5}mg + \frac{4}{5}Q)$

$\mu \geq \frac{\frac{3}{5}Q - \frac{4}{5}mg}{\frac{3}{5}mg + \frac{4}{5}Q}$

$\mu \geq \frac{3Q - 4mg}{3mg + 4Q}$

20 a)

$R(\nwarrow)$: $R - \frac{3}{\sqrt{13}}W = 0 \Rightarrow R = \frac{3}{\sqrt{13}}W$

$R(\nearrow)$: $F - W \times \frac{2}{\sqrt{13}} = 0 \Rightarrow F = \frac{2}{\sqrt{13}}W$

$\frac{2}{\sqrt{13}}W = \mu \times \frac{3}{\sqrt{13}}W \Rightarrow \mu = \frac{2}{3}$

20 b)

$R(\nwarrow\!\!\searrow)$: $R - \frac{6}{\sqrt{13}}W = 0 \Rightarrow R = \frac{6}{\sqrt{13}}W$

$F = \frac{2}{3} \times \frac{6}{\sqrt{13}}W =$

$R(\nearrow\!\!\swarrow)$: $\frac{4}{\sqrt{13}}W - 2W \times \frac{2}{\sqrt{13}} = 0$

The particle remains at rest.

(The friction has increased in the same proportion as the weight)

21 a)

On both:

$R(\nwarrow\!\!\searrow)$: $R - 2600 \times 10 \times \cos 10 = 0$

$\Rightarrow R = 26000 \cos 10$

$\Rightarrow F = 2600 \cos 10$

$R(\nearrow\!\!\swarrow)$: $6000 - 2600 \cos 10 - 2600 \times 10 \sin 10 = 2600a$

$\Rightarrow a = -0.414 \text{ ms}^{-2}$

Deceleration is 0.414 ms^{-2}

0.4 ms^{-2} (1 SF – AQA)

21 b)

On trailer:

$R(\nwarrow\!\!\searrow)$: $R - 400 \times 10 \times \cos 10 = 0$

$\Rightarrow R = 4000 \cos 10$

$\Rightarrow F = 400 \cos 10$

$R(\nearrow\!\!\swarrow)$: $T - 400 \cos 10 - 4000 \sin 10 = 400a$

$T = \frac{12000}{13}$

$T = 923$ N (3 SF)

$T = 900$ N (1 SF – AQA)

22 a)

$12 = 8 + 5a \Rightarrow a = 0.8 \text{ ms}^{-2}$

On both:

$R(\nwarrow\!\!\searrow)$: $R - 3000 \times 10 \times \cos 15 = 0$

$F = 0.15 \times 30000 \cos 15$

$R(\nearrow\!\!\swarrow)$: $D - 0.15 \times 30000 \cos 15 - 3000 \times 10 \times \sin 15 = 3000 \times 0.8$

$\Rightarrow D = 14511.23757\ldots$

$D = 14\,500$ N (3 SF)

$D = 10\,000$ N (1 SF – AQA)

22 b)

On trailer:

$R(\nwarrow\!\!\searrow)$: $R - 600 \times 10 \times \cos 15 = 0$

$\Rightarrow R = 6000\cos 15$

$\Rightarrow F = 900\cos 15$

$R(\nearrow)$: $T - 900\cos 15 - 6000\sin 15 = 600 \times 0.8$

$T = 2902.247514\ldots$

$T = 2900$ N (3 SF)

$T = 3000$ N (1 SF – AQA)

23)

$R(\nwarrow)$: $R - mg\cos 45 = 0 \Rightarrow R = \frac{\sqrt{2}}{2}mg$

$R(\nearrow)$: $P - F - mg\sin 45 = 0 \Rightarrow F = P - \frac{\sqrt{2}}{2}mg$

$P - \frac{\sqrt{2}}{2}mg = \mu(\frac{\sqrt{2}}{2}mg)$

$\Rightarrow \frac{2P}{\sqrt{2}mg} - 1 = \mu$

$\Rightarrow \mu = \frac{\sqrt{2}P}{mg} - 1$ as req

24)

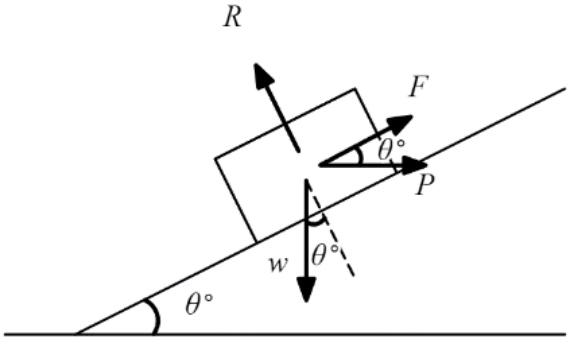
$R(\nwarrow)$: $R - mg\cos\theta = 0 \Rightarrow R = mg\cos\theta$

$F = \mu mg\cos\theta$

$R(\nearrow)$: $kmg - \mu mg\cos\theta - mg\sin\theta = ma$

$a = k - \mu g\cos\theta - g\sin\theta$

25 a)



25 b)

$\sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$

$R(\nwarrow)$: $R - \frac{24}{25}w - \frac{7}{25}P = 0 \Rightarrow R = \frac{24}{25}w + \frac{7}{25}P$

$R(\nearrow)$: $F + \frac{24}{25}P - \frac{7}{25}w = 0 \Rightarrow F = \frac{7}{25}w - \frac{24}{25}P$

$\frac{7}{25}w - \frac{24}{25}P = \mu(\frac{24}{25}w + \frac{7}{25}P)$

$7w - 24P = \mu(24w + 7P)$

$\mu = \frac{7w-24P}{24w+7P}$ as req.

25 c)

$R(\nwarrow)$: $R - w \times \frac{24}{25} = 0 \Rightarrow R = \frac{24}{25}w$

$F = \frac{24}{25}w\mu$

$R(\swarrow)$: $-\frac{24}{25}w\mu + w \times \frac{7}{25} = \frac{w}{g} \times \frac{1}{5}g$

$\Rightarrow -\frac{24}{25}w\mu = \frac{1}{5}w - \frac{7}{25}w \Rightarrow -24\mu = 5 - 7 \Rightarrow \mu = \frac{1}{12}$